Project Report On

AIDIE

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

in

Computer Science and Engineering

By

Noana Mary Charles (U2203168)

Pranathi K (U2203216)

Thomas George (U2203204)

Under the guidance of

Ms. Reshma M R

Department of Computer Science and Engineering
Rajagiri School of Engineering & Technology (Autonomous)
(Parent University: APJ Abdul Kalam Technological University)
Rajagiri Valley, Kakkanad, Kochi, 682039
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CERTIFICATE

This is to certify that the design report entitled ***"AIDIE"*** *is a bonafide record of the work done by* ***Noana Mary Charles (U2203168)***, ***Pranathi K (U2203216)***, ***Thomas George (U2203204)***, *submitted to the Rajagiri School of Engineering & Technology (RSET) (Autonomous) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in "Computer Science and Engineering" during the academic year 2025-2026.*

Project Guide
Ms. Reshma M R
Assistant Professor
Dept. of CSE
RSET

Project Co-ordinator
Ms. Jisha Mary Jose
Assistant Professor
Dept. of CSE
RSET

Head Of Department
Dr. Preetha K G
Professor
Dept. of CSE
RSET

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Last but not the least, We would like to express our sincere gratitude towards all other teachers and friends for their continuous support and constructive ideas.

Noana Mary Charles (U2203168)

Pranathi K (U2203216)

Thomas George (U2203204)

Abstract

The project entails the development of an AI-based assistive mobile app intended to help children with Attention Deficit Hyperactivity Disorder (ADHD) effectively deal with their attention span and emotional stability through an intelligent and empathetic digital interaction experience. The app is built with various AI-based features such as emotion detection, adaptive learning technology, and natural language processing to develop an intelligent and empathetic "buddy" that helps ADHD patients effectively deal with their condition. The app includes various intelligent and empathetic features such as game-based learning, emotional checks, and progress reports that help parents and educators track the progress and well-being of ADHD patients. The study aims to explore the development and evaluation of the app's effectiveness in promoting behavioral outcomes among ADHD patients with intelligent and empathetic interaction.

Contents

List of Abbreviations

Acronym - Expansion

AI - Artificial Intelligence

ML - Machine Learning

ADHD - Attention Deficit Hyperactivity Disorder

NLP - Natural Language Processing

UI/UX - User Interface/User Experience

UDL - Universal Design for Learning

STT - Speach To Text

TTS - Text To Speach

SDK - Software Development Kit

API - Application Programming Interface

DB - Database

RAM - Random Access Memory

GPU - Graphics Processing Unit

SSD - Solid State Drive

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Chapter 1

Introduction

1.1 Background

Attention Deficit Hyperactivity Disorder (ADHD) ranks as one of the most common neurodevelopmental disorders of childhood, as characterized by a pattern of inattention, hyperactivity and impulsiveness which affects the individual's day-to-day functioning as well as educational achievement. Conventional interventions for ADHD include behavioral therapy, a structured routine and as well as pharmacological interventions all of which require active participation of caregivers as well as educators.

In the modern world, advances in artificial intelligence have produced assistive technology that can provide children with ADHD with individualized support in real time. By identifying behavior and providing prompt treatments, this technology can be tailored to each person's needs. For this reason, mobile applications are thought to be a helpful tool. Through games or conversations in natural language, they may design dynamic worlds that captivate kids.

By offering conversational AI skills, emotional detection, and flexibility, an assistive AI-powered application for ADHD may help close the gap between therapy and everyday life. By helping the child with ADHD complete chores, receive direction in daily life, and develop coping skills for everyday situations, the application can serve as a virtual friend. It encourages a cooperative approach to treating ADHD by giving caregivers useful information on their child's development in addition to offering guidance to the child with ADHD.

1.2 Problem Definition

Children with ADHD often struggle with attention, emotional regulation, and completing everyday tasks, which affects their academic progress and social well-being. Existing tools provide some support but usually lack personalization, adaptability, and engaging features tailored to their needs. To address this, there is a need for a mobile app powered by assistive AI that offers adaptive reminders, gamified motivation, and real-time feedback. While also analyzing behavior patterns to provide personalized support for children and actionable insights for caregivers and teachers.

1.3 Scope and Motivation

The project's goal is to develop an AI-based mobile application for children with ADHD that would enhance their focus, emotional stability and capacity to complete activities by integrating various features like games, emotional feedback, reminders and progress tracking. By adjusting to the user's behavior, the project uses a variety of artificial intelligence techniques including machine learning and sentiment analysis to offer real-time advice. Additionally, the app is entertaining and easy to use for children with varying levels of attention.

The inability of children with ADHD to maintain focus and the ensuing demands on parents and teachers to provide them with ongoing direction are the driving forces behind this need. There is a need for intelligent solutions that provide sympathetic, adaptable and easily available instruction because conventional systems cannot provide this kind of real-time guidance outside of clinical settings. This is an attempt to lessen the stress on their caretakers and help kids develop self-regulation abilities.

1.4 Objectives

- To create an AI-powered mobile application that uses individualized interaction to help kids with ADHD manage their focus, emotions, and everyday responsibilities.
- to incorporate emotion detection technologies and adaptive learning algorithms that offer real-time, personalized feedback depending on the child's behavioral patterns.
- To incorporate gamified features that encourage motivation, engagement and sustained attention during learning or routine activities.
- To develop an intuitive and child-friendly user interface based on accessibility and UI/UX design principles suitable for ADHD users.
- To create a parent–teacher dashboard for tracking progress, monitoring emotional states and analyzing behavioral improvements over time.

1.5 Challenges

- Emotion Recognition Accuracy: Developing AI models with high accuracy in recognizing kids' emotions based on text, speech, or facial expressions can sometimes be challenging, as there are few labeled datasets for emotions.
- Personalization Complexity: Designing adaptive systems that respond effectively to each child's unique attention patterns and learning needs requires extensive behavioral data and fine-tuning.
- User Engagement: Maintaining consistent engagement among children with ADHD, who often have short attention spans—necessitates creative UI design and dynamic reward systems.
- Parental and Teacher Involvement: Real time information sharing while maintaining privacy and accessibility is challenging.
- Tech Accessibility: The AI features must be user friendly, device independent, offline accessible and reasonably priced in order to increase the app's accessibility.

- Evaluation and Validation: Long term evaluations including psychologists or educators may be required to assess the app's efficacy in improving behavior, attention and cognition.

1.6 Assumptions

For the real-time engagement the plan takes into consideration that the children who are going to use the app are going to have access to a device whether a tablet or a smartphone with a stable internet connection. In addition, the plan takes into consideration that the instructors and the parents are going to be part of this process as they are going to supervise the children's development through the connected interface. Moreover, the plan takes into consideration that the children who are diagnosed with ADHD are going to be able to focus on the interactive discourse of the software.

1.7 Societal and Industrial Relevance

Many children with ADHD throughout the world have their needs met by this AI-assisted program which enhances their focus, emotional stability and work habits to improve their quality of life. By accommodating various learning demands it lessens the burden of care and fosters inclusion in educational settings, so advancing social equality and neurodiversity.

From an industrial perspective, this idea ties in nicely with the rapidly expanding fields of educational technology and AI-assisted healthcare. By integrating state-of-the-art AI technology with user-centric designs, the project exemplifies innovation in AI-assisted digital therapeutic and child-centric educational applications. The program finds use in treating mental health and learning disorders, which are part of a sizable and expanding market.

1.8 Organization of the Report

The report is structured to guide the reader through the conceptual development and implementation of the proposed AI-based assistive learning system for children with ADHD. The *Literature Review* chapter discusses key research areas including AI-driven ADHD interventions, the integration of Universal Design for Learning (UDL) principles with artificial intelligence, the use of mobile learning apps for attention and behavior management, and neuroscience foundations of adaptive learning frameworks. These studies collectively establish the theoretical background and highlight gaps addressed by the project. The *System Design* chapter outlines the architecture and functionality of the proposed solution, explaining its modular structure comprising the child interface, parent-teacher dashboard, and AI-driven adaptive feedback engine. It also presents the inclusion of gamification logic, real-time progress tracking, and data security mechanisms, connecting research insights with practical design implementation.

1.9 Summary

This project develops an AI-powered mobile app to assist children with ADHD in managing focus, emotions, and daily routines through personalized, real-time support using emotion recognition, adaptive learning, and natural language processing. It features gamified motivation and progress tracking for parents and teachers to enhance attention, self-regulation, and engagement.

The app includes personalized reminders, emotion-based feedback, interactive learning modules, and an accessible UI tailored for ADHD needs. Motivated by challenges faced by children and caregivers, it aims to complement traditional therapy with intelligent digital support. Key objectives focus on AI integration, gamification, user-friendly design, and effectiveness evaluation.

Challenges include ensuring data privacy, accurate emotion detection, personalization, user engagement, caregiver integration, and accessibility. Assumptions involve device access, caregiver involvement, data availability, and ethical compliance. The app addresses societal needs by supporting children with ADHD and their caregivers while aligning with industry trends in AI-driven health and educational technologies, advancing personalized digital interventions for neurodiverse children.

Chapter 2

Literature Survey

Research on topics like AI-based therapy and interventions, the connection between UDL and AI, neuroscience based research in the context of UDL and the use of mobile technology in assistive technologies for students with ADHD has advanced quickly in the fields of educational technology, adaptive learning and ADHD support in recent years. The relationship between UDL and AI, the use of mobile technology in assistive technologies for students with ADHD, the use of neuroscience in the context of UDL and recent research in the areas of AI-based treatment and interventions in ADHD therapy are all critically examined in this section. This conversation will aid in comprehending the contributions and conclusions of the current study within the framework of the current research environment.

2.1 Literature Review

2.1.1 Integrating AI into ADHD Therapy

This study investigates the use of robotic assistants and artificial intelligence, namely Chat GPT 4o in the treatment of attention deficit hyperactivity disorder[1].It shows how, as compared to static or scripted digital media solutions, dynamic, real time contact with artificial intelligence can improve the degree of personalization, adaptability and involvement in therapy. High success rates in terms of tailored user engagement, extending attention span, reducing hyperactivity and favorable assessments from both caregivers and medical professionals have been demonstrated by technical studies and simulated clinical trials.

2.1.2 Universal Design for Learning and Artificial Intelligence

In this study, the possible link between the principles of Universal Design for Learning (UDL) and artificial intelligence (AI) is explored in the pursuit of promoting inclusion and autonomous learning in digital education settings[2]. In this regard, UDL is defined as the provision of multiple means of representation, engagement, and expression for the purpose of promoting accessibility among diverse learners. In the article, examples are provided of how AI can be utilized for the purpose of promoting diverse and inclusive learning opportunities in educational settings. The findings of the study are discussed in the context of promoting social justice in education settings.

2.1.3 Mobile Applications for Students with ADHD

An overview of the state of mobile applications created for kids with ADHD is the goal of this review. It highlights various features of these apps, including time and task management, cognitive training, self monitoring, mindfulness and investigates how they might help manage symptoms of ADHD as well as academic productivity and emotional regulation[3]. It acknowledges the benefits of assistive technology as well as the disadvantages related to potential abuse, inaccessibility and the requirement for customized app selection based on professional advice. The efficacy of these applications is supported by both quantitative and qualitative research with a focus on the necessity of mixed-methods approaches and their integration with other interventions.

2.1.4 UDL and the Learning Brain

This article links advances in neuroscience to Universal Design for Learning (UDL) offering a research-informed educational framework that addresses how the brain learns and innovatively supports diverse learners. UDL is grounded in understanding three interconnected neural networks in the brain that align with learning processes: the affective network, responsible for engagement and motivation (the "why" of learning); the recognition network, which processes perception and understanding (the "what" of learning); and the strategic network, involved in planning and expression (the "how" of learning). These networks correspond respectively to engagement, representation, and action/expression in the UDL framework. [4] It highlights that brain variability—not uniformity—is the dominant and defining feature of learning. Like fingerprints, no two brains are alike; each is shaped by genetics and experiences, creating unique neural pathways that evolve due to brain plasticity—the brain's remarkable ability to rewire itself through use and experience. This inherent neurodiversity means learners engage with, perceive, and act on information in multiple ways. UDL's proactive design acknowledges this variability by providing flexible, goal-driven learning environments that accommodate multiple pathways to learning success.

Paper	**Focus**	**Key Methods/Principles**	**Main Outcomes/Insights**
Mouratidis et al. (2024)	AI and robotics for ADHD therapy	Dynamic AI interaction, multimodal feedback, explainability, clinical simulation	Personalized, scalable therapy; improved attention and engagement; ethical + technical considerations
Al-Azawei & Badii (2023)	UDL + AI fostering inclusion	Multiple means of engagement, representation, expression, equity, ethics	AI-UDL synergy enables flexible, accessible, socially just education
Khasawneh et al. (2023)	ADHD-focused mobile apps	App features (task/time management, cognitive training), mixed-method analysis	Assistive technology aids learning and regulation; cautions on overuse; individualized, holistic approach advised
CAST (2022)	Neuroscience + UDL framework	Brain network alignment, learner variability, design for flexibility	Promotes expert, self-directed learning by targeting engagement, representation, expression

Table 2.1: Comparison of Key Literature on AI and Assistive Technologies for ADHD and Inclusive Education

Chapter 3

System Design

The overall system design for this ADHD assistive application makes use of machine learning (ML) and artificial intelligence (AI) techniques to provide children with ADHD with more individualized, flexible and responsive assistive support. For a more dynamic approach to treating ADHD symptoms for improvements in motivation, focus, and other areas, the overall architecture takes into account multimodal inputs like behavior, emotion and interaction.

3.1 Overall System Architecture

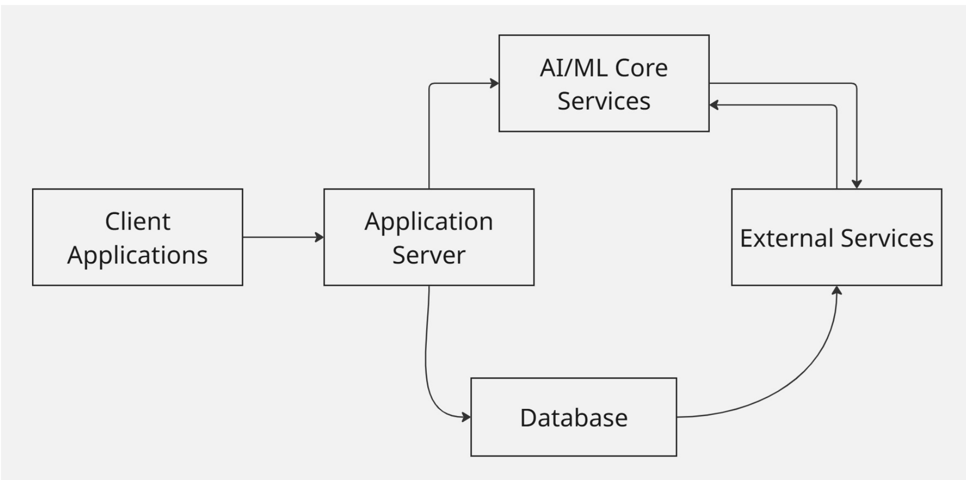


Figure 3.1: Overall System Architecture

- **Client Applications:** The dashboard application is intended for the parent or guardian, while the mobile application is intended for the youngster. By supplying input data these programs communicate with the system.

- **Application Server:** This manages user authentication as well as performs business logic for the application and serves as an API gateway. The client applications provide it with input data.
- **AI/ML Core Services:** The backend is connected to AI services and ML services. AI services comprise a conversational engine based on large language models and an adaptive learning/behavioral model. ML services provide personalized interaction and insights based on user data.
- **Database:** The system includes a database system that stores user profiles, activity data, progress logs, and content libraries. It is efficient in storing data and progress.
- **External Services:**The system includes external services such as Speech-to-Text (STT) and Text-to-Speech (TTS) services to support voice-based inputs.

The data flow is as follows, user interactions from the client applications are sent to the application server which manages authentication and logic. Relevant data is then passed to the AI/ML core services. These services may use external STT/TTS services for voice processing. The database stores and retrieves user data as well as supporting adaptive learning and progress monitoring. Parents and teachers can access insights and logs via the dashboard.

This model is not only secure and adaptive but also scalable and collaborative in nature, as it involves those who can provide the best support to ADHD patients.

3.2 Design Diagrams

The design diagrams in this section provide a comprehensive visual representation of the AIDIE system's modular architecture, illustrating client server interactions, AI driven workflows and data flows tailored for ADHD support in children. These figures from client applications and application servers to AI/ML core services, databases, external services, and component designs demonstrate real time emotion recognition integration (e.g., via FER2013 model), parental dashboards, and voice pipelines, ensuring scalability.

3.2.1 Module and its Details

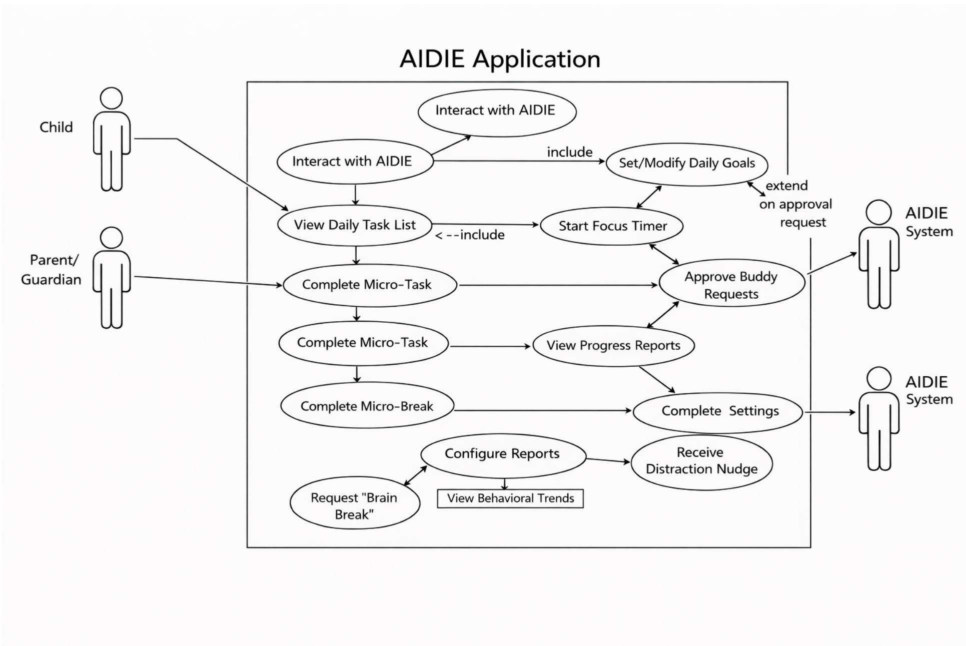


Figure 3.2: Client Applications

3.2.2 Client Application Module

- Child Interactions (left side) The child engages with AIDIE on a daily basis to ensure productivity. Key activities include setting daily goals or tasks (such as homework), starting a focus timer, completing micro-tasks (segmented activities), and asking for a brain break (ADHD-friendly breaks).

- Parent/Guardian Interactions (bottom-left) Parental supervision and intervention are involved. These actions include reviewing reports on progress, observing trends and configuration reports, or sending external requests (for example, modifying settings or adding tasks).

- AIDIE System Features (central and right side) Manages fundamental logic for managing Daily Tasks such as setting goals, starting timers, and completing micro-tasks and breaks. Generates reports on Progress, trends in behavior and configurations, and notifications for nudge avoidance distractions. Supports Responsive Settings such as repetitive configurations, viewing trends, and sending external system requests (for example, syncing with school systems).

- Overall Flow Arrows show bidirectional interactions—e.g., child completes tasks which then AIDIE generates reports and parent views them. It's designed for real-time engagement: children get nudges and timers for focus, while guardians monitor trends to customize support. This aligns perfectly with your facial emotion recognition project, where AIDIE could detect emotions to trigger breaks or nudges.

3.2.3 Application Server

This diagram illustrates the layered backend structure, with **Client Applications** communicating via HTTPS APIs to the server, which routes through controllers to services and storage.

Layer Breakdown

- **API Gateway**: Entry point for all client requests; handles routing, rate limiting, and security.

- **Controllers Layer**: Domain-specific handlers:
  - **ChildController**: Manages child-facing actions (e.g., task starts, brain breaks).
  - **TaskController**: Handles daily tasks, micro-tasks, and goal setting.
  - **ProgressController**: Tracks and reports progress, behavioral trends.
  - **ReminderController**: Sends nudges, distraction alerts.
  - **AIController**: Interfaces with ML for emotion detection, level calculations.

- **Middleware/Auth**: Cross-cutting concerns like authentication (e.g., JWT for parents), logging, and validation.

- **Firestore Client**: Connects to Firebase Firestore for real-time database (user data, sessions).
- **AI Client**: Communicates with external/internal AI Core Services (e.g., TensorFlow for facial emotion recognition).

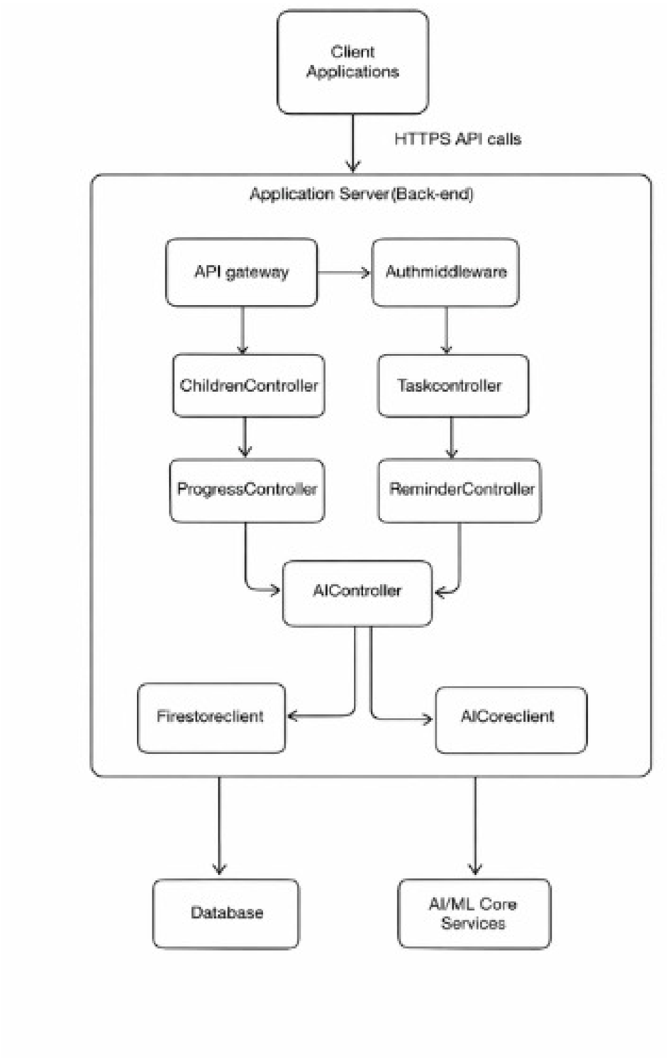


Figure 3.3: Application Server Module

3.2.4 AI/ML Core Services

This flowchart outlines the real-time **focus session workflow** integrated with facial emotion recognition for ADHD children in the AIDIE app.

Step-by-Step Process

- **Start Focus Session**: Initiate session; child begins work task.

- **Every 10s Camera Capture**: Continuous monitoring via front camera.
- **Emotion Recognition**: AI analyzes frame for emotions (e.g., frustration, distraction using fer2013-trained model).
- **Distraction/Frustration Detected?**
  - **No**: Check **Focus Lost?** → Continue Task.
  - **Yes**: Log event → Trigger **ADHD Intervention**.
- **ADHD Intervention Options**:
  - Voice nudge/help (e.g., "Take a deep breath").
  - Quick reset task.
  - Brain break request.
- **Intervention Effective?**
  - **No**: Restart timer, log, repeat recognition.
  - **Yes**: Log event → **Session Complete?**
- **Yes**: **Generate Report** (progress, trends).

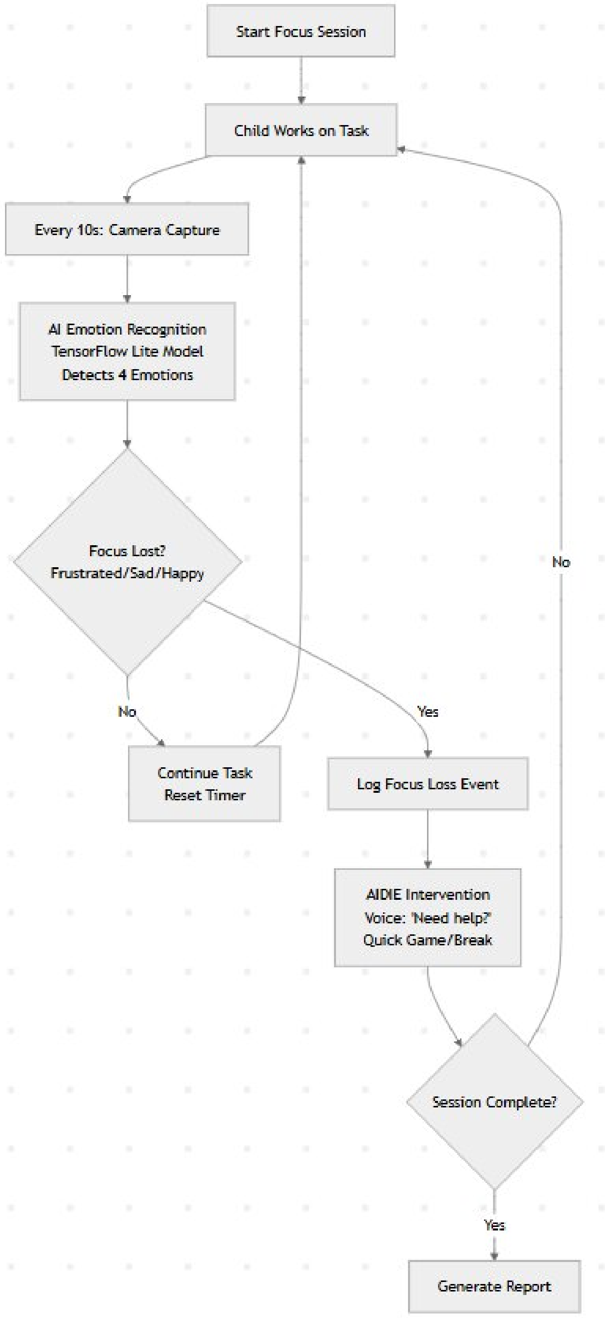


Figure 3.4: AI/ML Core Services Module

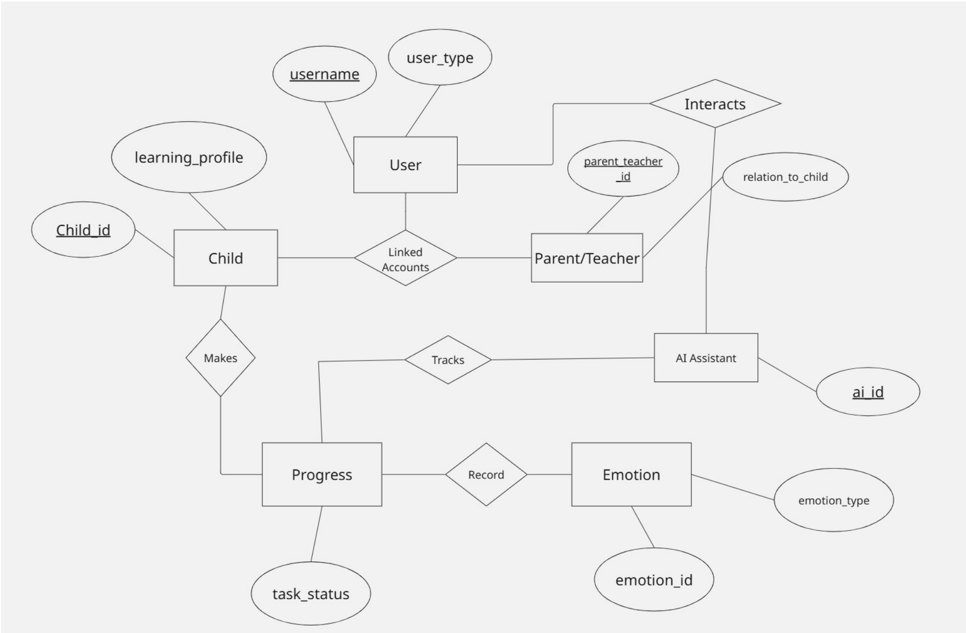


Figure 3.5: Database Module

3.2.5 Database

Entities: User, Child, Parent/Teacher, AI Assistant, Progress, Emotion.

Attributes: Each entity includes two major attributes, such as *child_id* and *learning_profile* for Child; *username* and *user_type* for User, and so on.

Relationships (Diamonds): Relationships like *Linked Accounts* (connecting Child and Parent/Teacher via User), *Interacts* (connecting User, Parent/Teacher, and AI Assistant), *Makes* (Child and Progress), *Tracks* (AI Assistant and Progress), and *Records* (Progress and Emotion) are indicated using diamonds.

Structure:

- User is central, linking to Parent/Teacher (who relates to children) and Child.
- Child is linked to Progress (shows tasks made by a child).
- Progress is connected to Emotion (logs emotional state per task) and tracked by AI Assistant.
- AI Assistant has *Interacts* and *Tracks* relationships, linking User activities and Progress records.

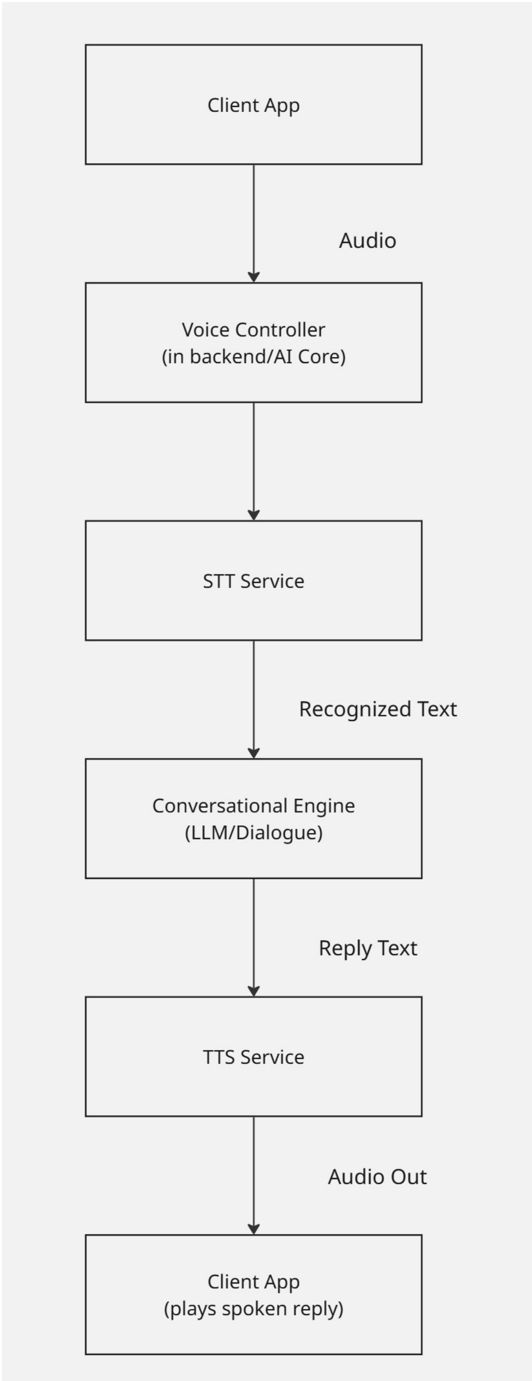


Figure 3.6: AI/ML Core Services Module

3.2.6 External Services

This diagram shows the complete voice-based conversational flow from client app speech input to audio output, integrating external AI services.

Pipeline Components:

- Client App (Speech Input): Child/parent speaks into microphone.
- Voice Controller (Backend AI Core): Receives raw audio, handles session management.
- STT Service: Speech-to-Text conversion (e.g., Google Cloud Speech, Azure Speech).
- Recognized Text: Converts speech to structured text.
- Conversational Engine (LLM/Dialog Manager): Processes intent (e.g., "I need a break") using LLM like GPT or custom dialog model.
- Reply Text: Generated natural language response.
- TTS Service: Text-to-Speech synthesis (e.g., Google TTS, Amazon Polly).
- Client App (Audio Out): Plays synthesized speech response to user.

This modular structure enables efficient data flow between user interfaces, behavioral analysis, adaptive learning systems, parental controls, and voice services, collectively enhancing support for children with ADHD.

3.2.7 Sequence Diagram

The figure illustrates how the big task is divided and completed by the child with the assistance of AI through the AI helper app. This is how the AI helper app works:

- The child types in the big task in the mobile app, like "Do math homework."
- It displays "Thinking..." on the screen so that the child can see that it is working.
- The server receives the big task and user context.
- It also receives any additional context that might be required for the big task.

- Based on what the user has given, it divides the big task into smaller tasks that can be completed easily
- It saves the micro-task and reminder in the database and sends it to the mobile app.
- It displays the micro-task and reminder to the child, like "Let's start with your math book."
- It also schedules the reminder so that the child remains motivated.

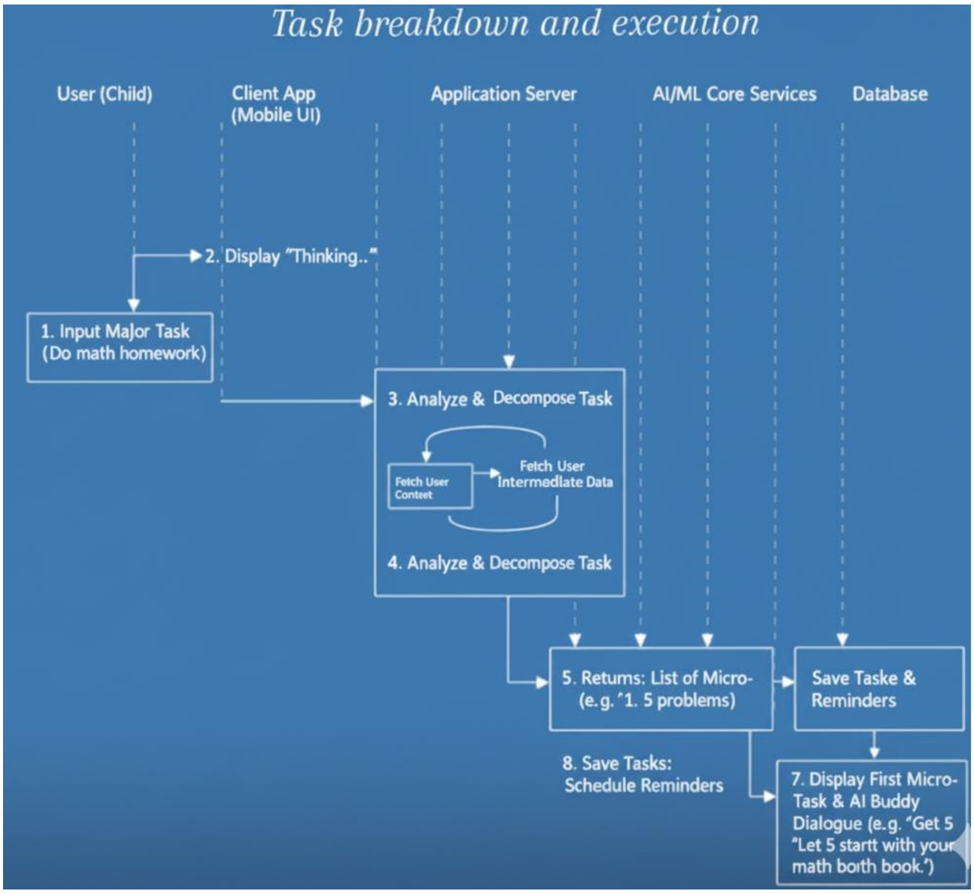


Figure 3.7: Sequence Diagram

3.3 Component Design

The component design of the AI assistive application defines the internal structure and interaction between different software modules. Each component is designed to perform specific functions, communicate through defined interfaces, and maintain modularity for scalability and upgradability.

Major Components

1. **Client Application:** Handles user interaction through mobile and web interfaces. It captures emotional check-ins, task engagement data, and progress.
2. **Application Backend:** Manages authentication, session tracking, and communication between the client, AI core, and database.
3. **AI/ML Core:** Responsible for natural language processing, adaptive learning, behavioral analysis, and emotional response generation.
4. **Database:** Stores user data, activity logs, task configurations, and content library.
5. **External Services:** Integrates speech-to-text (STT) and text-to-speech (TTS) for voice-based interaction.
6. **Parent/Guardian Dashboard:** Provides analytics, reports, task customization, and notifications for monitoring child progress.

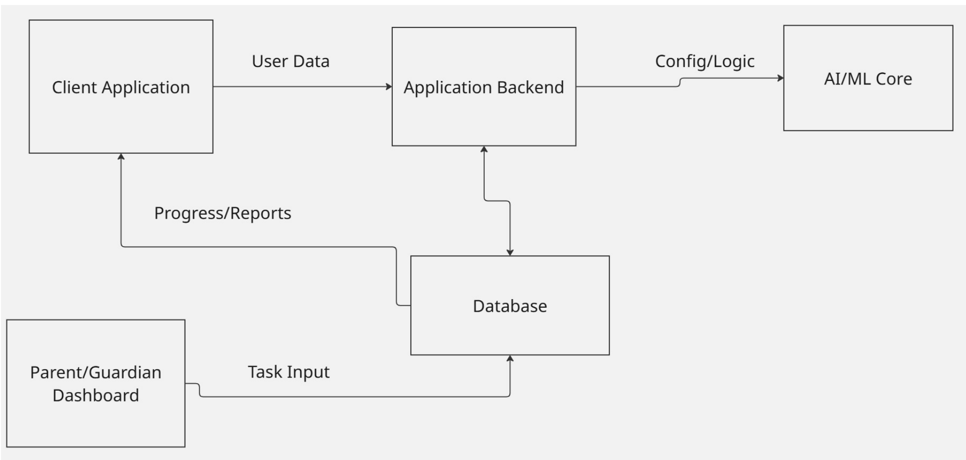


Figure 3.8: Component Design

3.4 Software and Hardware Requirements

3.4.1 Software Requirements

Client Application (Frontend)

- Android 8.0 (API 26)+; Android 12+ optimal for TFLite/ML Kit features.faceonlive+2
- Support for Web access (optional, React.js, Angular)
- UI library for accessible and engaging interfaces
- Google Play Services for ML Kit (face detection).
- CameraX 1.3+ and TensorFlow Lite 2.14+ libraries.
- Speech processing SDKs (Google Speech-to-Text, Text-to-Speech APIs)
- Machine Learning libraries for on-device prediction (TensorFlow Lite, Core ML)

Backend Services

- Cloud platform (AWS, Azure, Google Cloud) supporting scalable API hosting
- Relational databases
- ML model hosting platforms or custom model-serving microservices
- APIs for dialogue generation and behavioral analysis (TensorFlow Serving, FastAPI)
- User data analytics and logging mechanisms
- Notification services (Firebase Cloud Messaging, OneSignal)

3.4.2 Hardware Requirements

Client Devices

- Smartphones or tablets with microphone and speaker for voice interaction
- Minimum 2GB RAM, 1.5 GHz processor for smooth app performance
- Camera access (optional) for emotion recognition features

- Internet connectivity for cloud-based AI processing and data sync

This setup will allow development of a scalable, secure, and user-responsive system while supporting all needed AI-driven features and rich interaction modes suitable for children with ADHD. Adjustments depend on your deployment scale and specific components used.

3.5 Dataset Identified

3.5.1 FER 2013 Dataset

The FER-2013 dataset is a benchmark for facial expression recognition, containing 35,887 grayscale facial images labeled into 7 basic emotions: anger, disgust, fear, happiness, sadness, surprise, and neutral.

Dataset Characteristics

- **Size**: 28,709 training, 3,589 validation, 3,589 test images
- **Resolution**: 48×48 pixels (grayscale)
- **Source**: Google image search crawl + manual labeling via crowdsourcing
- **Classes**: 7 emotions (imbalanced distribution)
- **Challenges**: Occlusions, lighting variations, pose changes, low resolution

3.6 Module Division and Work responsibilities

Noana Mary Charles

Main Focus: Server, Firebase setup, APIs, and data management.

Responsibilities:

- Set up Node.js + Express backend.
- Configure Firebase Firestore and Firebase Authentication.
- Implement REST API routes:
  - /api/users → create, get, update user profiles.

  - /api/games → manage curriculum-based games.
  - /api/progress → track progress, points, and rewards.

- Manage .env variables and deploy backend (optional: on Render or Firebase Functions).
- Write basic unit tests (optional).

Thomas George

Main Focus: App screens, navigation, and interaction design.

Responsibilities:

- Design UI wireframes for Login/Signup, Student Dashboard, Game selection, Progress tracker and rewards.
- Build frontend with React Native (mobile) or React.js (web).
- Integrate frontend with backend API.
- Design accessible and ADHD-friendly UX (color coding, focus timers, simple navigation).
- Implement reward visuals (stars, badges).

Pranathi K

Main Focus: Educational games, progress tracking, and gamification logic.

Responsibilities:

- Design curriculum-based mini-games (Math, Reading, Focus games).
- Connect game results with progress API.
- Implement backend logic (points, levels, achievements).
- Create reward system (unlock badges, XP levels).
- Handle adaptive difficulty based on user performance.
- Develop data visualization for progress charts.

3.7 Expected Outputs

The AI-assisted ADHD app aims to produce measurable improvements in several key areas:

- **Improved Focus and Attention:** Detects lapses using behavioral and neurological signals, and adapts task difficulty and scheduling dynamically to sustain engagement.
- **Enhanced Task Completion and Time Management:** Utilizes rapid triage and adaptive time-boxing to help users prioritize tasks effectively, break down complex work, and stay on schedule with reminders.
- **Increased Motivation and Reduced Frustration:** Provides motivational nudges, reward scheduling, and next-step prompts that lower cognitive load and maintain user interest and confidence.
- **Personalized Learning Experience:** Tailors content pacing and difficulty via data-driven insights from educational and emotional engagement datasets, fostering better retention and mastery.
- **Real-Time Emotional Support:** Recognizes emotional states such as boredom or overwhelm through monitored physiological signals and behavior, triggering timely supportive interventions.

Collectively, these outputs contribute to improved academic performance, daily functioning, and quality of life for children with ADHD by providing adaptive, personalized, and context-aware support.

3.8 Project Timeline

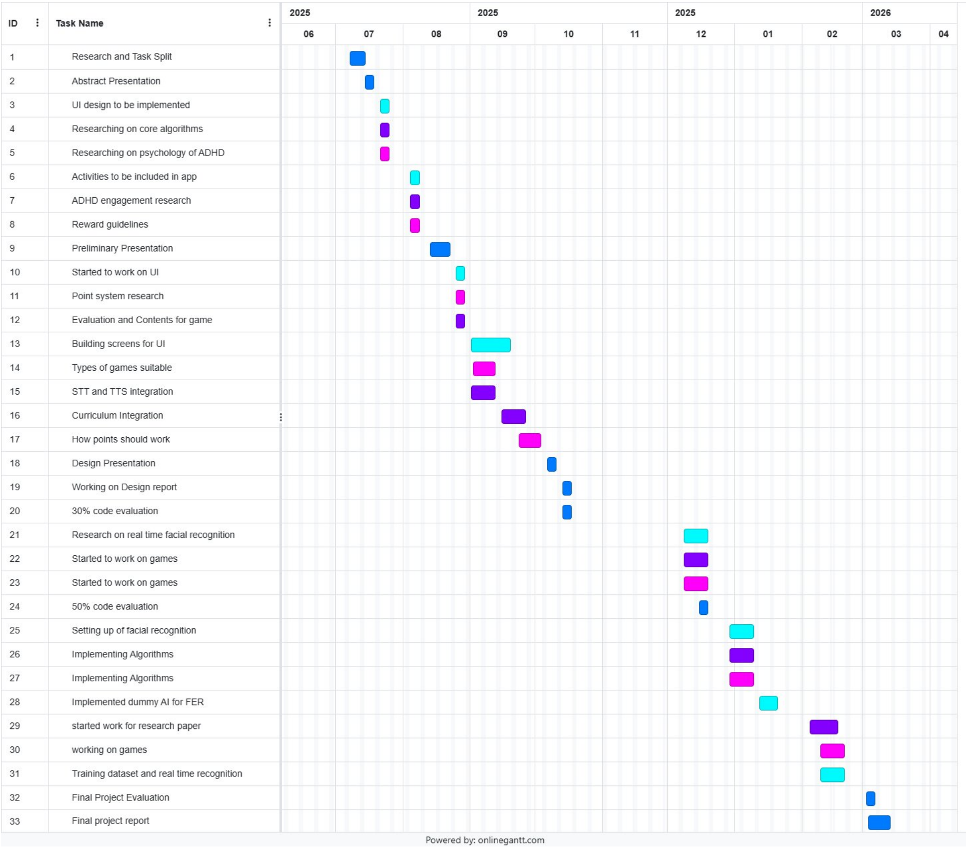


Figure 3.9: Project Timeline

Blue: All, Turquoise: Thomas, Purple: Noana, Pink: Pranathi

3.9 Summary

This chapter outlines the architecture and components of the ADHD assistive application, which integrates mobile interfaces, AI-powered intervention logic and secure backend infrastructure. The system operates in a modular fashion where the child and parent interact the app that capture behavioral, emotional and learning data; all data is processed by a backend application server that communicates with advanced AI/ML services for real-time feedback and adaptive suggestions. Core modules include Aidie for conversational and motivational support, task management with gamified micro-tasks and reminders, adaptive content recommendation and a comprehensive parent dashboard. The design leverages external services for seamless voice interaction and robust data storage for accessibility. Algorithms for task triage, time-boxing, engagement detection and reward scheduling to personalize the learning journey as well as utilizing public datasets for training and benchmarking. The app aims to enhance focus, task completion, motivation and emotional well-being in children with ADHD, while providing clear progress feedback to caregivers through thoughtfully designed interfaces.

Chapter 4

Results and Discussions

This chapter presents the results of the developed mobile application, focusing on the user interface design and the performance of the facial emotion recognition module. The discussion highlights how these components work together to support children with ADHD in monitoring their emotions and staying on task.

4.1 Overview of the Application UI

The final user interface was designed with a child-friendly layout, large buttons, and minimal on-screen distractions. The home screen provides quick access to the three main features: Mood Meter, Mindfulness, and Task Time, along with a points and rewards panel to motivate regular use (Figure 4.5). The color palette and visuals were intentionally kept simple so that children can recognize each feature without having to read long labels. Additional educational games are also accessible from the home screen, including a literacy game for rearranging jumbled letters to form words, a math game using voice input for answering problems aloud, and a fun action game called Save the Bunny, all earning points to encourage engagement.

4.1.1 Educational Games

The Literacy game challenges children to rearrange jumbled letters into correct words like “CAT” or “ROCKET” by dragging tiles, building vocabulary and focus (Figure 4.1).
The Math Voice Master game displays problems like 12×7 and requires speaking the answer aloud for voice recognition verification, reinforcing math skills interactively (Figure 4.2).
The Save the Bunny game is a fun platformer where players guide a bunny past obstacles to earn points, promoting quick thinking and hand-eye coordination (Figures 4.3 and 4.4).

4.2 Task Time Interface

The Task Time screen allows the child to focus on a single activity for a short, manageable period (Figure 4.6). The interface provides a text field for naming the task (for example, "Homework") and three preset time options of 5, 10, and 15 minutes so that the child does not need to configure complex timers. Once a duration is selected, a large countdown timer and progress bar are displayed, along with clearly distinguishable Start and Reset buttons.

4.3 Mood Meter and Mindfulness Screens

The screen for the Mood Meter simply says "How do you feel right now?" and presents the child with small buttons of common emojis that express how someone might be feeling (Figure 4.7). Selecting the mood earns the child "mood points" and logs the mood so that it can be referred to later to track the child's moods throughout the day or session. The exercise is brief on purpose so that it can be completed even if the child is in an emotional state. The Mindfulness screen takes the mood selection and provides the child with short

activities that match the mood they were in (Figure 4.8). The screen provides the option for the child to select between breathing, body scan, and the "5-4-3-2-1" exercise. Each activity has a description that is easy to read, a timer and progress bar assist the child in completing the exercise. A second mood check is also included on the screen at the bottom that says "How do you feel now?"

4.4 Facial Emotion Recognition Results

The AI Emotion Detection screen enables real-time or photo-based classification of a child's facial expression into discrete categories such as happy, sad, and frustrated (Figures 4.9–4.12). For each prediction, the app overlays a coloured label at the top of the image (for example, a green "Happy" label with an accompanying emoji) so that the output is immediately understandable. This output can be used either as feedback during a study session.

4.5 Parent/Teacher Dashboard

A Parent/Teacher dashboard provides a summary of how the child has used the application and how he or she is progressing(Figure 4.13). This dashboard is available after a separate login.Providing a quick overview without requiring the user to sift through raw data. The 'Engagement Levels' dashboard (Figure 4.14) shows daily and weekly activities such as total study time (32 minutes today), activities completed (5 this week), distribution of time spent on studying, games, mindfulness, and tasks. There is a weekly bar chart to display how consistent these activities are. Tips such as 'Engagement looks healthy. Keep sessions short, positive, and consistent' are provided to help parents or teachers improve the child's usage. Goals such as '20-30 minutes of focused activity on at least 5 days a week' help set targets. Academic Performance shows a summary of the performance achieved as well as grades for individual subjects(Figure 4.15). It highlights areas of strength (e.g Maths at 85%) and areas that require improvement (e.g Science at 72%, 2 missing lab assignments). It also shows recently missed work that needs to be completed along with due dates. Skill Development shows areas like Focus (Level 4, improving), Organisation, Task Completion, and Emotional Regulation. These are displayed as a summary of sessions completed and improvements achieved (for example, Task Completion: 5 sessions, 40 minutes this week).(Figure 4.16). The Mood Meter view for parents aggregates recent checks (e.g., mostly happy today, 3 checks; weekly breakdowns: Happy 40%, Calm 24%, Angry 14%) and offers notes like "Adjust schedule around tough days" (Figure 4.17). The Update Schedule screen lets caregivers edit time blocks (e.g., 04:00–04:30 PM for Homework/Reading) or add new ones, ensuring alignment with school routines.

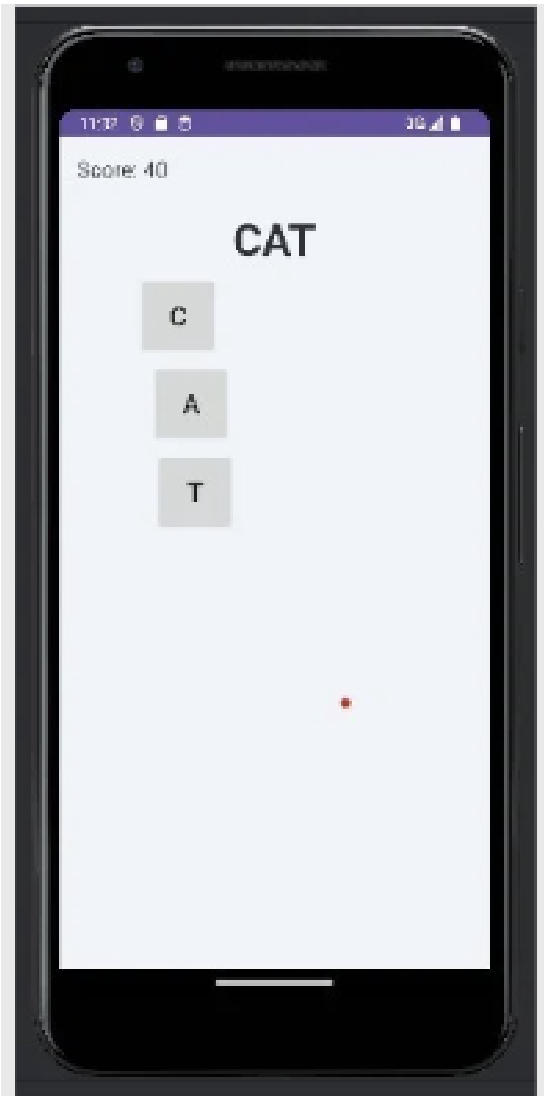
4.6 Integration of UI and Emotion Detection

The main achievement of the project is the combined use of UI components and the emotion recognition module in one process. On the home screen, there are "Check My Focus" and "Complete Task." These options help the child complete a task, possibly checking their facial expression during the process, and finally assessing their emotions after the session. If the system detects a negative facial expression or frustration during the process, the child may be asked to open the mindfulness screen and complete a short activity before continuing with the task. The parent/teacher dashboard joins data from

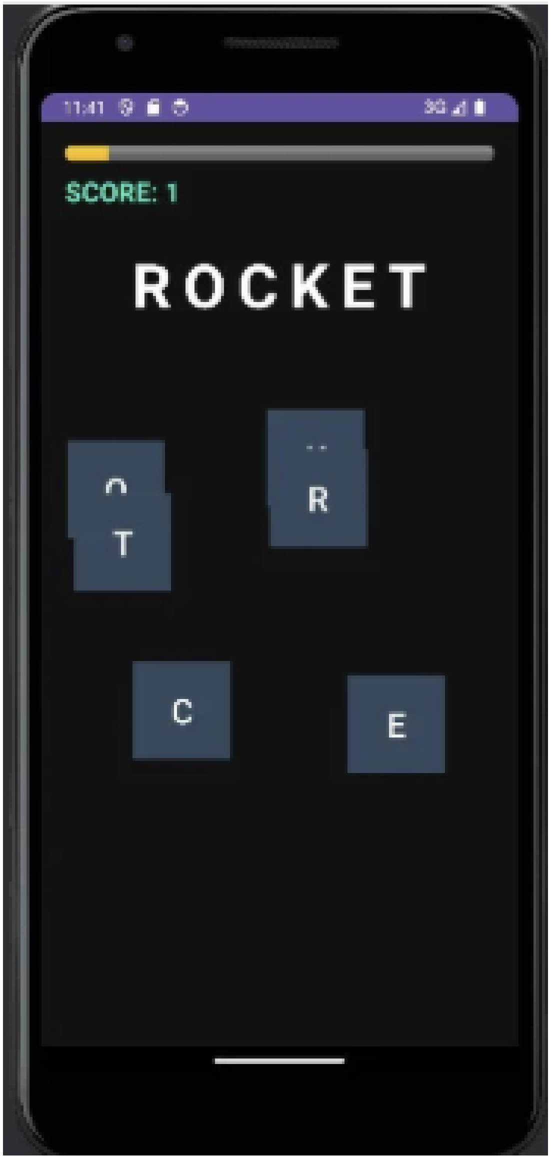
these child facing features which allows caregivers to manage emotion logs, task completion rates and AI-detected expressions with academic and skill metrics. This demonstrates the potential of using AI in emotion recognition with simple ideas in UX design such as using a timer, emojis, and reward points. It also highlights some important considerations in using the project, such as the need for proper lighting and camera positioning and the need to use AI results as a prompt rather than a judgment of a child's emotions.

4.7 Summary

The AIDIE mobile application successfully integrates a child-friendly UI with real-time facial emotion recognition to support ADHD management, featuring accessible screens for Task Time, Mood Meter, Mindfulness, and engaging educational games like Literacy game and Save the Bunny. Performance highlights include accurate emotion classification (happy, sad, focused, frustrated) via FER2013-trained models, comprehensive parent dashboards tracking engagement, academics, skills, and moods for proactive interventions. This holistic approach demonstrates AI's potential in assistive education, with considerations for lighting/camera setup emphasizing its role as a supportive tool rather than definitive judgment.



(a) Literacy game: Unscrambling “CAT”



(b) Literacy game: Unscrambling “ROCKET”

Figure 4.1: Literacy game screens showing jumbled letter rearrangement.

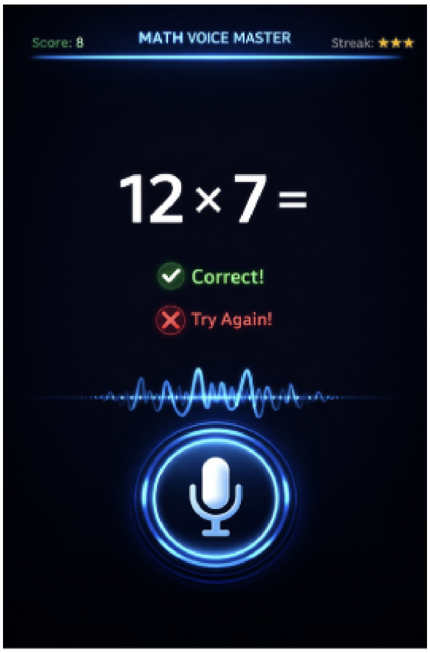
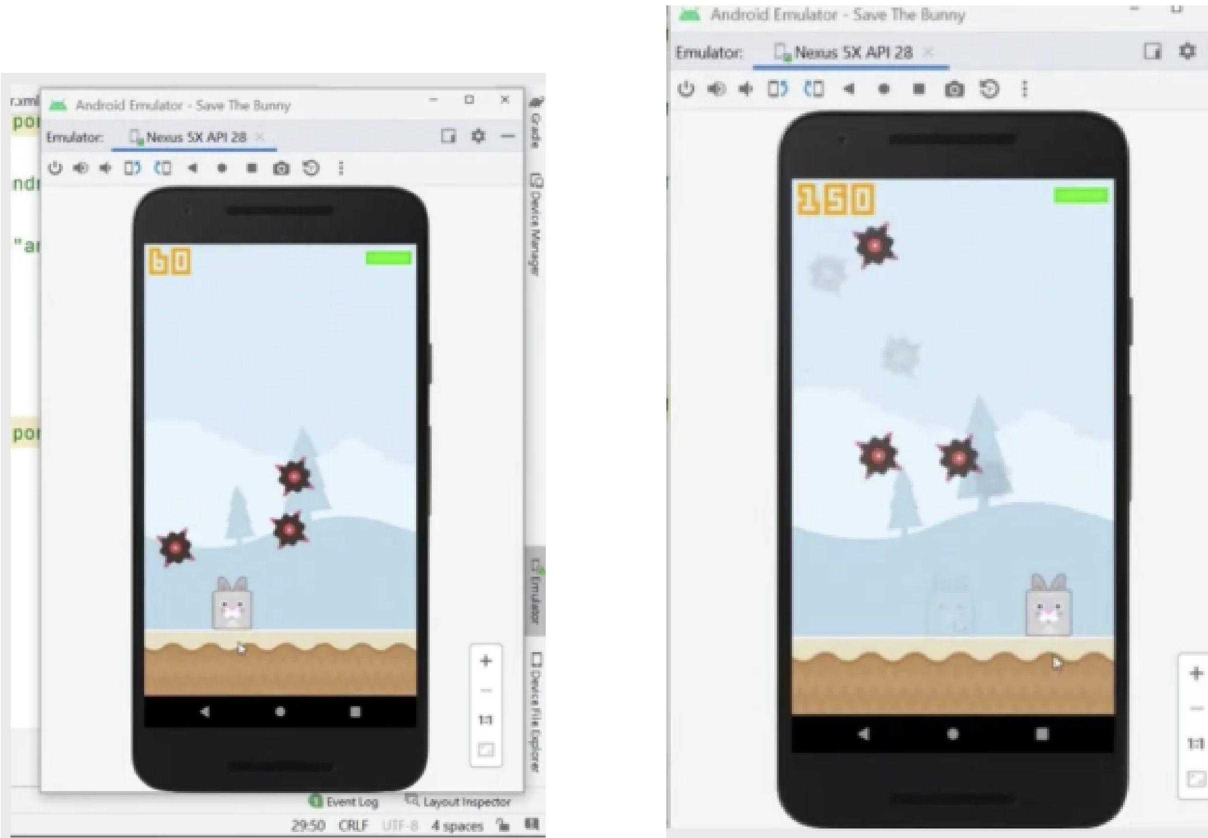


Figure 4.2: Math Voice Master game with voice input for multiplication problems.



(a) In-game action with score and bunny.

(b) Points earned screen.

Figure 4.3: Save the Bunny game: Gameplay and rewards screens.

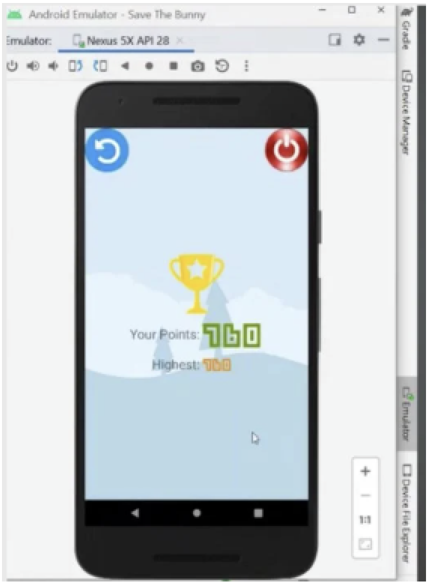


Figure 4.4: Save the Bunny high score achievement.

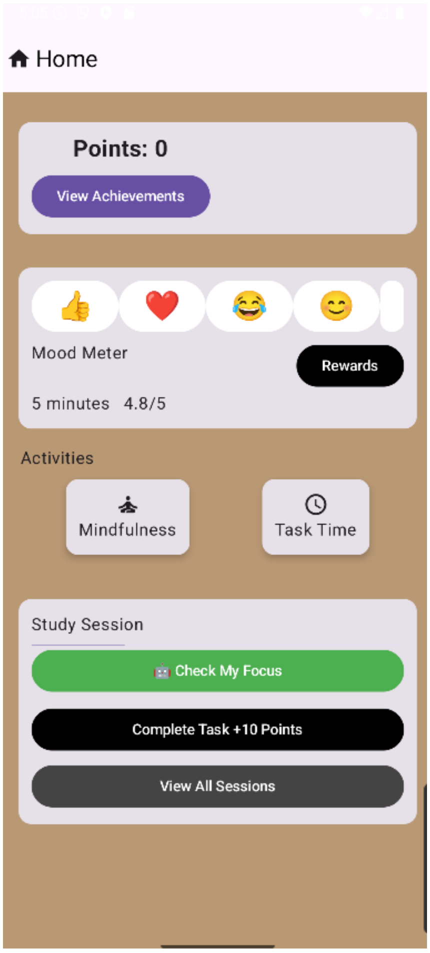


Figure 4.5: Home screen of the application showing points, recent moods, and access to core activities.

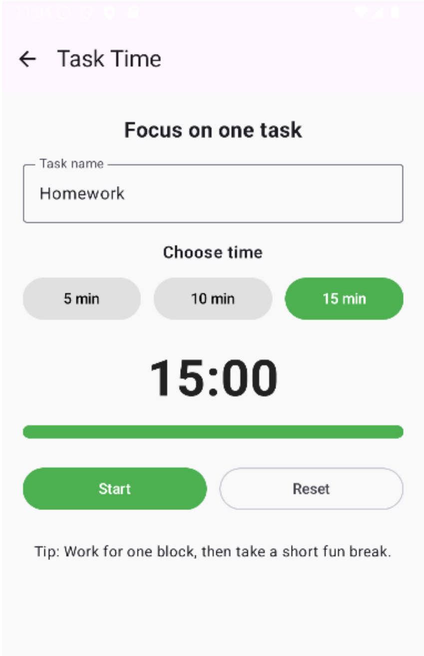


Figure 4.6: Task Time interface with task name, preset durations, and countdown timer.

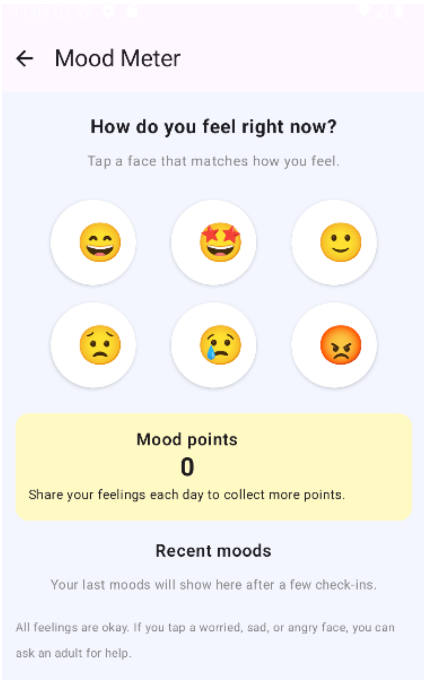


Figure 4.7: Mood Meter screen for quick self-report of emotional state.

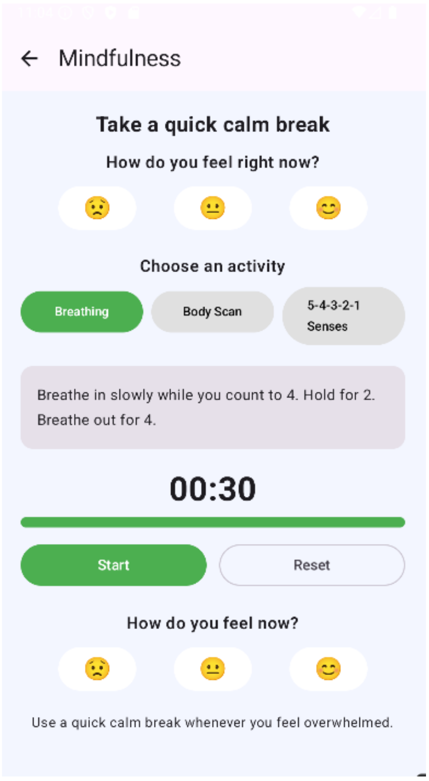


Figure 4.8: Mindfulness screen offering short calming activities with an integrated timer.

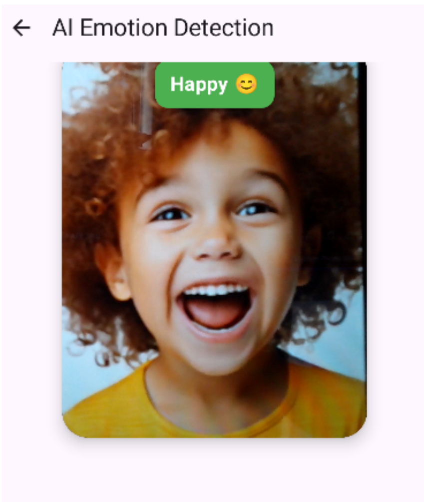


Figure 4.9: AI Emotion Detection screen classifying a happy expression.

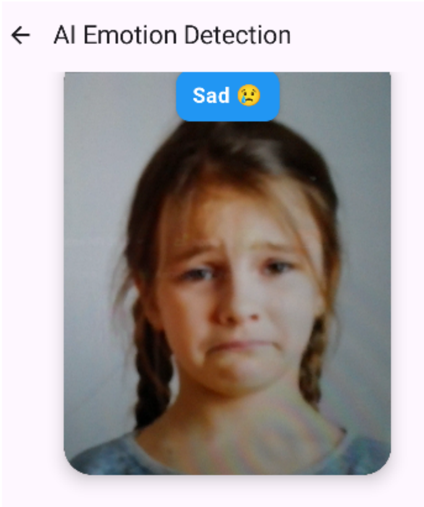


Figure 4.10: AI Emotion Detection screen classifying a sad expression.

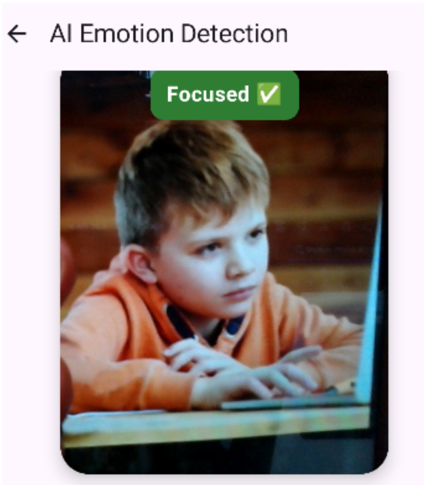


Figure 4.11: AI Emotion Detection screen classifying a focused expression.

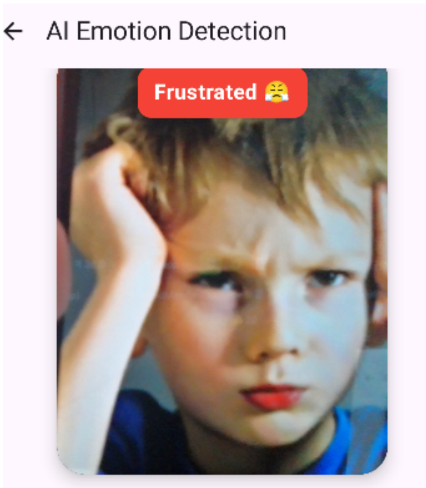


Figure 4.12: AI Emotion Detection screen classifying a frustrated expression.

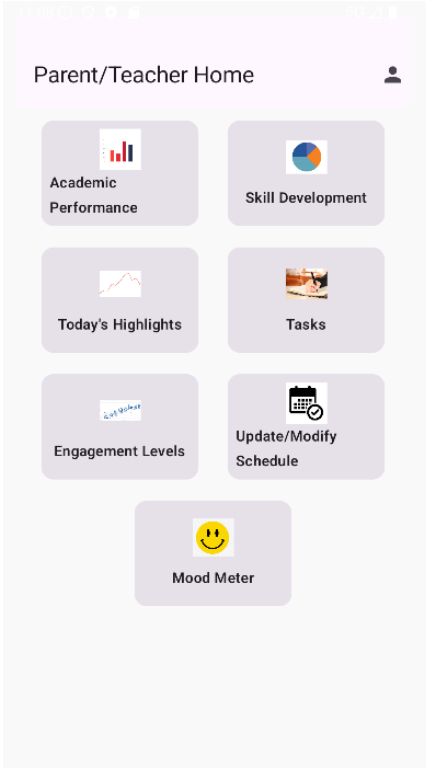


Figure 4.13: Parent/Teacher dashboard home screen for key insights.

Figure 4.14: Engagement Levels screen showing time allocation and weekly trends.

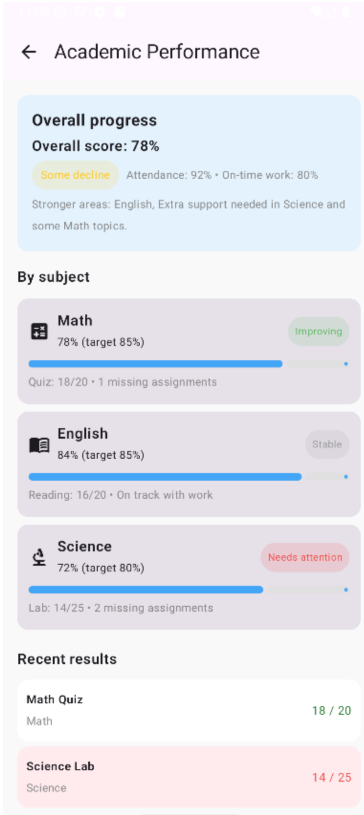


Figure 4.15: Academic Performance overview with subject breakdowns.

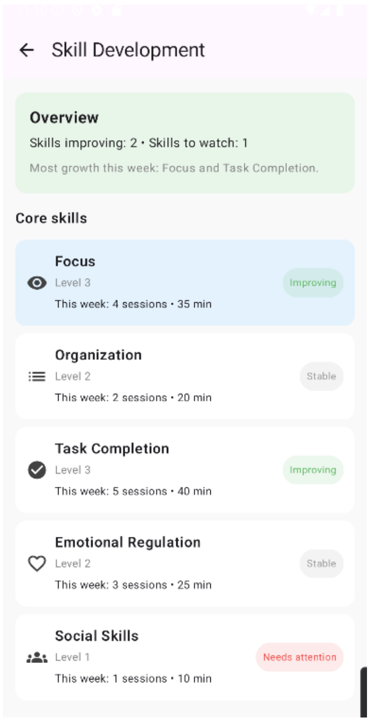


Figure 4.16: Skill Development tracking core areas like focus and emotional regulation.

Figure 4.17: Parent view of Mood Meter with weekly emotion breakdowns.

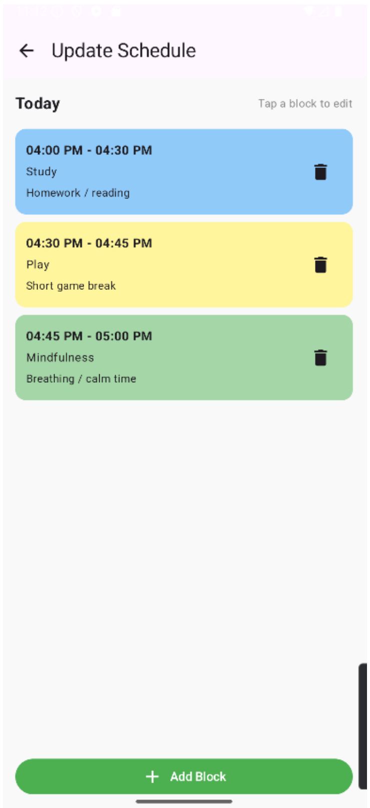


Figure 4.18: Update Schedule screen for customising daily activities.

Chapter 5

Conclusion

5.1 Conclusion

This project successfully developed a comprehensive mobile application that integrates facial emotion recognition with child-friendly UI elements to support children with ADHD. The app combines mood tracking, mindfulness exercises, task timers, educational games, and parent dashboards into a cohesive system that promotes emotional awareness, focus, and academic engagement through gamification and real-time feedback.

5.2 Project Limitations

While functional, the current implementation has limitations typical of a student project scope. The emotion recognition model, trained on datasets like FER2013, achieves acceptable accuracy under controlled lighting but may struggle with diverse skin tones, facial angles, or poor camera conditions common in real-world use. The app lacks longitudinal user testing with actual ADHD children and families, relying instead on design principles and emulator-based validation. Educational games provide engagement but need adaptive difficulty scaling based on individual performance. Parent dashboard analytics are descriptive rather than predictive, missing advanced correlations between emotions, task completion, and academic metrics. Finally, the app operates offline for core features but could benefit from cloud syncing for multi-device access and data backup.

5.3 Future Scope and Recommendations

Several enhancements would elevate the application for production deployment and clinical validation:

- **Model Improvements**: Fine-tune the emotion recognition model with ADHD-specific datasets capturing subtle expressions (e.g., frustration during tasks). Implement multi-modal detection combining facial analysis with voice tone from math games or physiological signals via wearables.

- **Advanced Features**: Add adaptive task scheduling using emotion trends (e.g., suggest mindfulness before tasks when frustration is detected). Integrate AR elements in games for immersive learning. Enable peer matching for collaborative challenges based on skill levels.

- **Clinical Integration**: Conduct randomized controlled trials measuring outcomes like on-task behavior, academic grades, and parent stress reduction. Partner with schools for integration into IEPs (Individualized Education Programs).

- **Technical Enhancements**: Deploy on cloud infrastructure for real-time analytics and AI model updates. Add NLP for voice command task creation and sentiment analysis in journal entries. Support multi-language localization for broader accessibility.

- **Accessibility**: Implement voice-over for visually impaired users, haptic feedback for timer alerts, and customizable color themes for color-blind users.

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Appendix A: Presentation

FINAL PRESENTATION ON AIDIE

3RD MARCH 2026

BY:
NOANA MARY (U2203168)
THOMAS GEORGE (U2203204)
PRANATHI K (U2203216)

GUIDE: MS.RESHMA MR

CONTENTS

01

PROBLEM DEFINITON

- Children with ADHD often struggle with attention, emotional regulation, and completing everyday tasks, which affects their academic progress and social well-being.
- Existing tools provide some support but usually lack personalization, adaptability, and engaging features tailored to their needs. To address this, there is a need for a mobile app powered by assistive AI that offers adaptive reminders, gamified motivation, and real-time feedback.
- While also analyzing behavior patterns to provide personalized support for children and actionable insights for caregivers and teachers.

02

PROJECT OBJECTIVES

- To design a mobile application that provides personalized task management through adaptive reminders and step-by-step guidance suited to children's attention spans.
- To enhance engagement and motivation by incorporating gamification and positive reinforcement strategies.
- To support emotional regulation by integrating simple mindfulness activities, calming prompts, or mood check-ins.
- To ensure the app is child-friendly, safe, and accessible, with intuitive interfaces tailored to the needs of children with ADHD.

03

PURPOSE AND NEED

Purpose
- Build a child friendly app for ADHD emotional support
- Provide encouragement to help children complete tasks
- Create calmer learning experiences for ADHD children and families

Need
- No child-friendly tool exists for real time emotion tracking
- Present tools lack adaptability

04

LITERATURE SURVEY

1. Challenges in ADHD Management for Children
Current research highlights that children with ADHD face significant hurdles in:
- Executive Functioning: Struggling with attention, organizing daily routines, and completing tasks.
- Emotional Regulation: Difficulty in managing moods, which impacts both academic progress and social integration.
- Support Gaps: A critical need exists for tools that offer real-time tracking and encouragement for both the children and their support networks (families and teachers) **[Chen and Zhang, 2021][DuPaul and Stoner, 2011]**.

2. Limitations of Existing Digital Tools
A survey of current assistive technology reveals that while tools exist, they often fail to meet the specific requirements of neurodivergent users because:
- Lack of Personalization: Many tools are one-size-fits-all and do not adapt to an individual child's attention span or unique needs.
- Engagement Deficits: There is a notable lack of gamified motivation and positive reinforcement strategies tailored for children.
- Static Nature: Most existing tools lack real-time adaptability and the ability to track emotions effectively in a child-friendly manner. **[Holmberg and Persson, 2020][Dovis et al., 2015].**

05

LITERATURE SURVEY

3. Emerging Assistive AI and mHealth Solutions

The proposed method aligns with modern literature advocating for "empathetic" and "intelligent" assistive systems that utilize:

- Affective Computing: Using AI for emotion recognition—identifying states like frustration or focus—to provide immediate, non-invasive support.
- Adaptive Interventions: Implementing systems that analyze behavioral patterns to provide personalized task recommendations and "nudge" users when distractions occur.
- Multimodal Interaction: Integrating voice-based services like Speech-to-Text (STT) and Text-to-Speech (TTS) to reduce cognitive load and improve accessibility for children who may struggle with traditional text-based interfaces. **[Yadegaridehkordi et al., 2019][Gao et al., 2024]**

4. Collaborative Ecosystems for Caregivers

Literature suggests that successful intervention requires a "closed-loop" system where data is not only used for the child but also translated for caregivers. The proposed architecture supports this by:

- Providing Actionable Insights: Generating behavioral trends and progress reports for parents, teachers, and therapists.
- Synchronized Support: Utilizing a centralized database accessed via a secure backend to coordinate care between the child's mobile app and a parent/teacher dashboard. **[Shah and Bhat, 2024].**

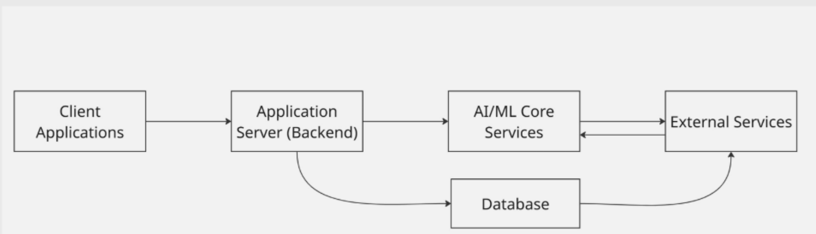
05

PROPOSED METHOD

- The proposed method develops an AI-powered mobile application using real-time facial emotion recognition to support children with ADHD.
- Gamified activities—breaking frustrated sessions into micro-tasks, extending focused learning modules.
- The parent/teacher dashboard with emotion trend analytics, task completion insights.

06

ARCHITECTURE DIAGRAM



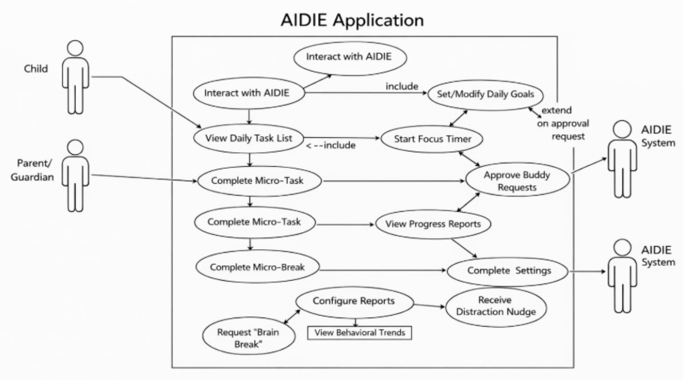
07

MODULE DESCRIPTION

- Client Applications: Interfaces for children and parents/teachers; handles UI.
- Application Server (Backend): Acts as API gateway with authentication routes requests, manages sessions, and coordinates between clients and core services.
- AI/ML Core Services: Performs emotion recognition (happy, sad, focused, frustrated), behavioral analysis, adaptive interventions, and task recommendations.
- External Services: Integrates speech-to-text (STT) and text-to-speech (TTS) for voice interactions with the app
- Database: Centralized storage for user profiles, emotion histories, tasks, progress, and session data; accessed only via backend.

08

MODULE WISE DIAGRAMS

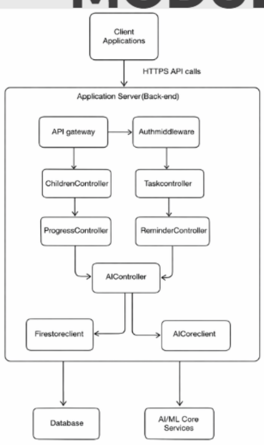


Client Applications

- Mobile app (child UI) and web/parent dashboard.
- Collect user input: tasks, mood, answers, voice,and show outputs: reminders, micro-tasks, rewards.
- Communicate only with the application server, they never talk directly to the database or AI core.

9

MODULE WISE DIAGRAMS



Application Server (Backend)
Handles user authentication (Firebase/Auth) and acts as an API gateway.

Exposes REST endpoints for children, t asks, progress, reminders and AI features (breakTask, nextActions,sessionConfig, insights).

Contains business logic to validate requests, enforce roles (parent vs child),and orchestrate calls to AI/ML Core Services and the Database.

10

MODULE WISE DIAGRAMS

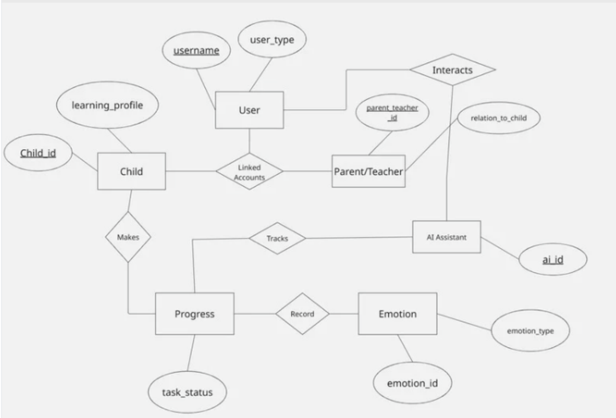
AI/ML Core Services

Flow Explanation

1. Start Focus Session → Child begins working on a task
2. Child Works on Task → Normal monitoring phase
3. Every 10s: Camera Capture → Takes face photo automatically
4. AI Emotion Recognition → Your model analyzes 4 emotions (focused vs frustrated/sad/happy)
5. Focus Lost? → Decision point:
   - No → Child continues working, timer resets
   - Yes → Triggers help sequence
6. Log Focus Loss Event → Records distraction for reports
7. AI Buddy Intervention → Friendly voice prompt ("Need help?") or quick game
8. Session Complete? → Either loops back or ends with summary report

10

MODULE WISE DIAGRAMS



Database

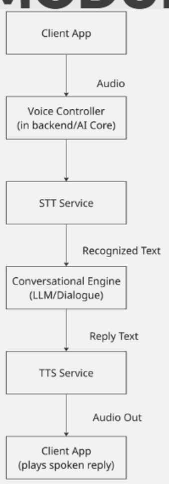
Central store for user profiles, child settings, tasks, assignments, progress logs, mood entries, and content library (activities, prompts, rewards).

Supports both transactional reads/writes from the backend (e.g., completeTask) and analytics queries from the AI/ML core (e.g., patterns over weeks).

Ensures persistence, security, and consistency of all ADHD-related data.

11

MODULE WISE DIAGRAMS



External Services

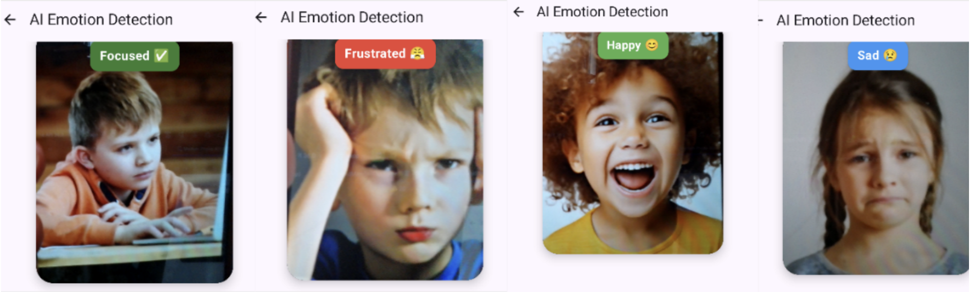
Speech-to-Text (STT) converts the child's or parent's voice into text for the conversational engine or task input.

Text-to-Speech (TTS) turns AI or system responses into child-friendly speech output.

Integrated by the AI/ML or backend layer, not directly by the client apps, so keys and privacy are controlled centrally.

12

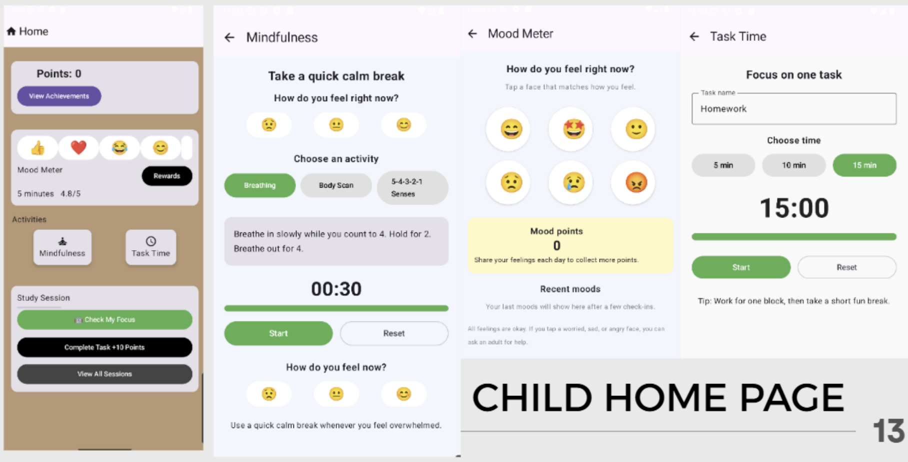
RESULTS AND OUTPUTS



- THE VARIOUS EMOTIONS BEING DETECTED
- DETECTED BY SCANNING AREA OF THE EYES AND MOUTH

13

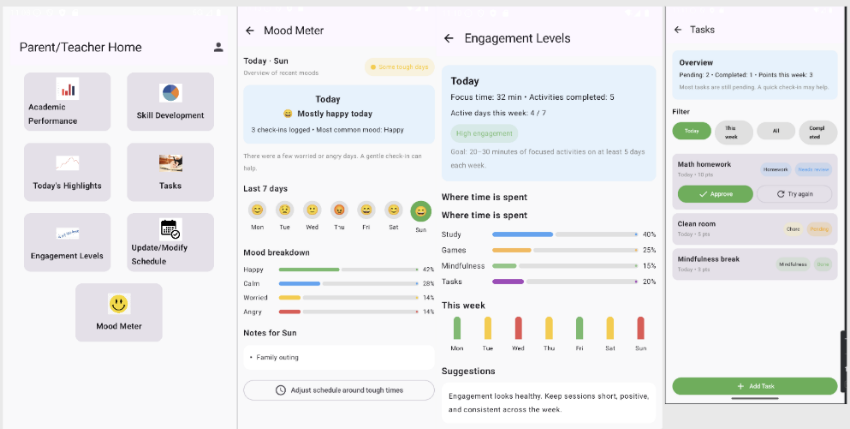
RESULTS AND OUTPUTS



CHILD HOME PAGE

13

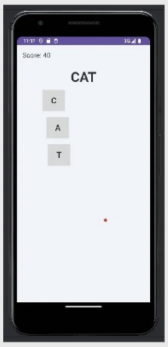
RESULTS AND OUTPUTS

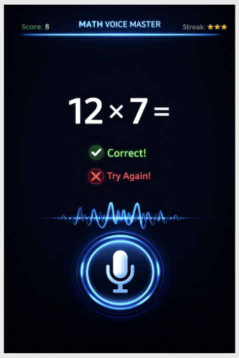


PARENT/TEACHER DASHBOARD

13

RESULTS AND OUTPUTS

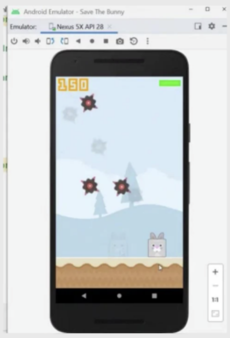


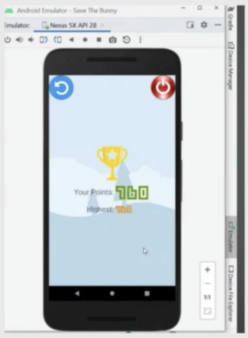


LITERACY GAME MATH VOICE MASTER

13

RESULTS AND OUTPUTS





SAVE THE BUNNY GAME

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ASSUMPTIONS

- Children have access to a tablet or computer.
- Parents/teachers will supervise usage.
- Consistent daily use builds self-regulation skills, not replacing–therapy or medication.
- Children use the app in familiar settings (home, school) with minimal distractions during short sessions.

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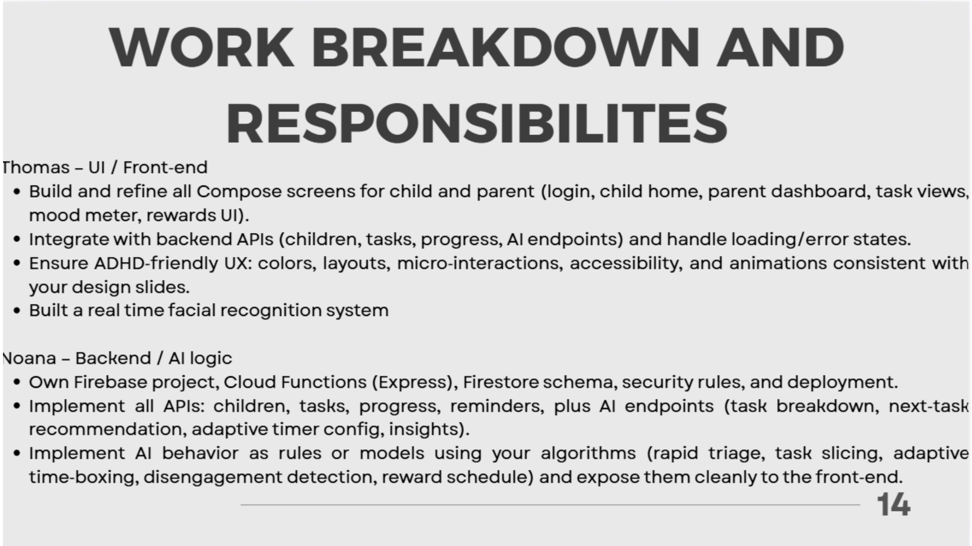
WORK BREAKDOWN AND RESPONSIBILITES
Thomas – UI / Front-end
• Build and refine all Compose screens for child and parent (login, child home, parent dashboard, task views, mood meter, rewards UI).
• Integrate with backend APIs (children, tasks, progress, AI endpoints) and handle loading/error states.
• Ensure ADHD-friendly UX: colors, layouts, micro-interactions, accessibility, and animations consistent with your design slides.
• Built a real time facial recognition system
Ioana – Backend / AI logic
• Own Firebase project, Cloud Functions (Express), Firestore schema, security rules, and deployment.
• Implement all APIs: children, tasks, progress, reminders, plus AI endpoints (task breakdown, next-task recommendation, adaptive timer config, insights).
• Implement AI behavior as rules or models using your algorithms (rapid triage, task slicing, adaptive time-boxing, disengagement detection, reward schedule) and expose them cleanly to the front-end.
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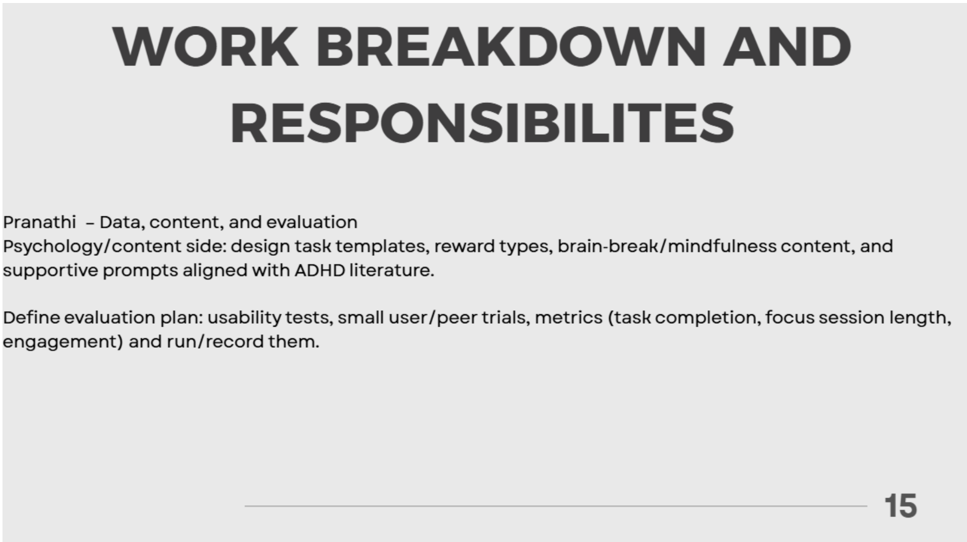
WORK BREAKDOWN AND RESPONSIBILITES
Pranathi – Data, content, and evaluation
Psychology/content side: design task templates, reward types, brain-break/mindfulness content, and supportive prompts aligned with ADHD literature.
Define evaluation plan: usability tests, small user/peer trials, metrics (task completion, focus session length, engagement) and run/record them.
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HARDWARE AND SOFTWARE REQUIREMENTS
Hardware Requirements
• Processor: ARM-based 1.5 GHz quad-core or better (e.g., Snapdragon 600 series+ or equivalent) for smooth 30fps.
• RAM: Minimum 2-3 GB total (256 MB+ free for app); 4 GB+ recommended for multitasking.
• Storage: 100 MB+ free space for app and caches.
• Camera: Front-facing 5 MP+ (640x480 resolution minimum) for reliable face capture.
• Display: 720p+ screen for child-friendly UI; touch-enabled
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HARDWARE AND SOFTWARE REQUIREMENTS

Software Requirements
Client (Android Device):

- Android 8.0 (API 26)+; Android 12+ optimal for TFLite/ML Kit features.faceonlive+2
- Google Play Services for ML Kit (face detection).
- CameraX 1.3+ and TensorFlow Lite 2.14+ libraries.

Dataset used

- FER2013: Primary dataset for initial TensorFlow Lite emotion model. Mapped 4-class system (happy, sad, focused, frustrated)

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GANTT CHART

Blue: All
Turquoise: Thomas
Purple: Noana
Pink: Pranathi

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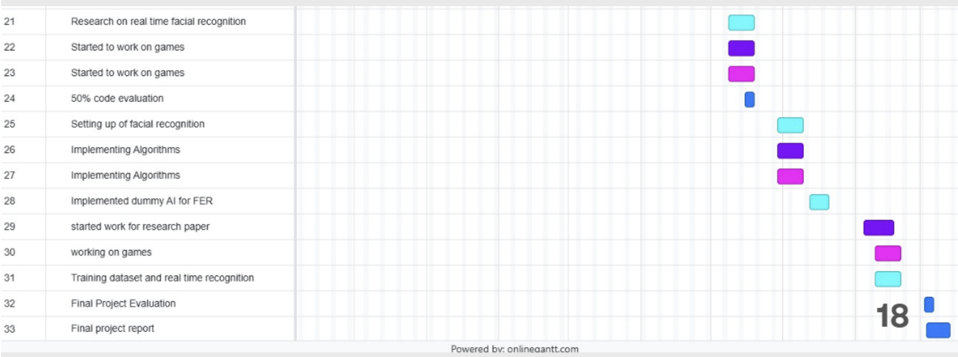
GANTT CHART

Blue: All Turquoise: Thomas Purple: Noana Pink: Pranathi



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RISK AND CHALLENGES

- Reward fatigue - children lose interest in static points system.
- Measuring long-term behavioral impact requires clinical studies
- Short attention spans
- Children engage consistently with gamified rewards and AI interventions despite short attention spans.

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RISK AND CHALLENGES

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- Measuring long-term behavioral impact requires clinical studies
- Short attention spans
- Children engage consistently with gamified rewards and AI interventions despite short attention spans.

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CONCLUSION

- This project proposes an AI-powered assistive mobile application designed to support children with ADHD in managing daily tasks and improving focus. By combining adaptive reminders, gamification, and ,the system provides personalized support tailored to individual needs.
- The application also empowers caregivers and teachers with actionable insights, contributing to improved academic performance and overall well-being. Future enhancements can include advanced AI models and broader accessibility features.

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THANK YOU

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Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes

Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

1. Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

2. Problem analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

3. Design/development of solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

4. Conduct investigations of complex problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.

5. Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

6. The Engineer and the world: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

7. Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.

8. Individual and Collaborative Team Work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

9. Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

10. Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

11. Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

Programme Specific Outcomes (PSO)

A graduate of the Computer Science and Engineering Program will demonstrate:

PSO1: Computer Science Specific Skills
The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas by understanding the core principles and concepts of computer science and thereby engage in national grand challenges.

PSO2: Programming and Software Development Skills
The ability to acquire programming efficiency by designing algorithms and applying standard practices in software project development to deliver quality software products meeting the demands of the industry.

PSO3: Professional Skills
The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs thereby evolving as an eminent researcher and entrepreneur.

Course Outcomes (CO)

Course Outcome 1: Model and solve real world problems by applying knowledge across domains (Cognitive knowledge level: Apply).

Course Outcome 2: Develop products, processes or technologies for sustainable and socially relevant applications (Cognitive knowledge level: Apply).

Course Outcome 3: Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks (Cognitive knowledge level: Apply).

Course Outcome 4: Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms (Cognitive knowledge level: Apply).

Course Outcome 5: Identify technology/research gaps and propose innovative/creative solutions (Cognitive knowledge level: Analyze).

Course Outcome 6: Organize and communicate technical and scientific findings effectively in written and oral forms (Cognitive knowledge level: Apply).

Appendix C: CO-PO-PSO Mapping

COURSE OUTCOMES

SNO	DESCRIPTION	BLOOM'S TAXONOMY LEVEL
1	Model and solve real world problems by applying knowledge across domains.	Apply
2	Develop products, processes or technologies for sustainable and socially relevant application.	Apply
3	Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks.	Apply
4	Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms.	Apply
5	Identify technology/research gaps and propose innovative/creative solutions.	Analyze
6	Organize and communicate technical and scientific findings effectively in written and oral forms.	Apply

MAPPING COURSE OUTCOMES (COs) – PROGRAM OUTCOMES (POs) AND PROGRAM SPECIFIC OUTCOMES (PSOs)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2	2	1		2	2	1	1	1	1	1	2	2
CO2	2	2	2		2	3	2	1	1	1	1	2	2	
CO3								3	2	2	1			3
CO4		1	2		3	2	3	2	2	3	1		2	2
CO5	2	3	3	1	2					1	3	3		
CO6			2			2	2	2	3	1	1			3

JUSTIFICATIONS FOR CO-PO MAPPING

MAPPING	LOW / MEDIUM / HIGH	JUSTIFICATION
101003/ CS822U.CO1- PO1	Medium	Applies engineering knowledge, mathematics, and computing fundamentals to model and solve real-world problems.
101003/ CS822U.CO1- PO2	Medium	Requires identification, formulation and analysis of engineering problems before developing solutions.
101003/ CS822U.CO1- PO3	Medium	Involves designing systems or solutions to address identified engineering needs.
101003/ CS822U.CO1- PO4	Low	Limited investigation or experimentation may be required to validate the developed models or solutions.
101003/ CS822U.CO1- PO6	Medium	Solutions developed may influence societal, environmental, or sustainability aspects of engineering applications.
101003/ CS822U.CO1- PO7	Medium	Encourages consideration of ethical and professional responsibilities while developing solutions.
101003/ CS822U.CO1- PO8	Low	Some level of teamwork may be required while working on modeling or project tasks.
101003/ CS822U.CO1- PO9	Low	Requires communication of results and findings through reports or presentations.
101003/ CS822U.CO1- PO10	Low	Involves basic planning and execution of tasks while implementing solutions.
101003/ CS822U.CO1- PO11	Low	Encourages independent learning and adapting new knowledge to solve real-world problems.
101003/ CS822U.CO1- PSO1	Medium	Uses core computer science principles to analyze problems and design appropriate technical solutions.

101003/ CS822U.CO1- PSO2	Medium	Involves algorithm design and application of software development practices to implement solutions.
101003/ CS822U.CO1- PSO3	Medium	Applies computer science fundamentals in research or innovative problem solving for societal needs.
101003/ CS822U.CO2- PO1	Medium	Utilizes engineering knowledge to design and develop products or systems.
101003/ CS822U.CO2- PO2	Medium	Requires analysis of user needs and technological feasibility before development.
101003/ CS822U.CO2- PO3	Medium	Focuses on design and development of solutions addressing real-world requirements.
101003/ CS822U.CO2- PO5	Medium	Requires use of modern engineering and IT tools for system or product development.
101003/ CS822U.CO2- PO6	High	Emphasizes sustainability and societal impact during solution development.
101003/ CS822U.CO2- PO7	Medium	Ethical and professional responsibilities are considered when developing socially relevant technologies.
101003/ CS822U.CO2- PO8	Low	Development activities may require collaboration among team members.
101003/ CS822U.CO2- PO9	Low	Involves communicating design ideas and development outcomes.
101003/ CS822U.CO2- PO10	Low	Requires basic planning and coordination during development tasks.
101003/ CS822U.CO2- PO11	Low	Encourages continuous learning of emerging technologies while building solutions.
101003/ CS822U.CO2- PSO1	Medium	Applies computer science fundamentals to design technically feasible solutions.
101003/ CS822U.CO2- PSO2	Medium	Uses programming and software development practices to implement solutions.
101003/ CS822U.CO3- PO8	High	Strongly relates to functioning effectively as a member or leader in multidisciplinary teams.

101003/ CS822U.CO3- PO9	Medium	Requires effective communication among team members during collaboration.
101003/ CS822U.CO3- PO10	Medium	Team leadership involves project coordination and task management.
101003/ CS822U.CO3- PO11	Low	Team interaction supports learning from peers and adapting new approaches.
101003/ CS822U.CO3- PSO3	High	Professional collaboration and teamwork support innovative computer science solutions.
101003/ CS822U.CO4- PO2	Low	Requires analysis and planning before executing project tasks.
101003/ CS822U.CO4- PO3	Medium	Involves designing and implementing project solutions based on planned activities.
101003/ CS822U.CO4- PO5	High	Requires effective use of modern tools and resources during project execution.
101003/ CS822U.CO4- PO6	Medium	Considers societal and environmental impact while implementing solutions.
101003/ CS822U.CO4- PO7	High	Emphasizes adherence to ethical practices and professional responsibilities.
101003/ CS822U.CO4- PO8	Medium	Task execution may involve teamwork and coordination among team members.
101003/ CS822U.CO4- PO9	Medium	Requires communication of progress, outcomes and documentation.
101003/ CS822U.CO4- PO10	High	Focuses on project management including scheduling, resource allocation and task execution.
101003/ CS822U.CO4- PO11	Low	Encourages adaptability and continuous learning during project implementation.
101003/ CS822U.CO4- PSO2	Medium	Uses software development practices to execute and manage project tasks.
101003/ CS822U.CO4- PSO3	Medium	Applies computer science knowledge in practical project development environments.

101003/ CS822U.CO5- PO1	Medium	Uses engineering knowledge to understand technological limitations and opportunities.
101003/ CS822U.CO5- PO2	High	Requires deep problem analysis and identification of research gaps.
101003/ CS822U.CO5- PO3	High	Proposes innovative design solutions to address identified gaps.
101003/ CS822U.CO5- PO4	Low	May involve limited investigation or validation of proposed ideas.
101003/ CS822U.CO5- PO5	Medium	Uses appropriate tools and techniques to analyze and develop innovative solutions.
101003/ CS822U.CO5- PO10	Low	Requires minimal planning while proposing new solution approaches.
101003/ CS822U.CO5- PO11	High	Strongly promotes independent learning and exploration of emerging technologies.
101003/ CS822U.CO5- PSO1	High	Applies computer science fundamentals to identify and solve complex technical problems.
101003/ CS822U.CO6- PO3	Medium	Requires presentation of design concepts and developed solutions.
101003/ CS822U.CO6- PO6	Medium	Communication may include discussion of societal or environmental implications.
101003/ CS822U.CO6- PO7	Medium	Requires responsible and ethical communication of technical results.
101003/ CS822U.CO6- PO8	Medium	Communication supports collaboration within team environments.
101003/ CS822U.CO6- PO9	High	Strongly relates to effective written and oral communication of technical findings.
101003/ CS822U.CO6- PO10	Low	Requires structured reporting and documentation in project management contexts.
101003/ CS822U.CO6- PO11	Low	Encourages continuous improvement in communication and presentation skills.
101003/ CS822U.CO6- PSO3	High	Communication of innovative computer science solutions and research outcomes.